

Product Requirements Document

AI-Based Mining Compliance, Inspection, and Risk Management Mobile Application

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1. Executive Summary

This product is a mobile-first compliance and inspection platform for mining organizations. It enables administrators, mine managers, inspectors, regulatory authorities, and corporate management teams to manage mining structures, monitor regulatory obligations, conduct field inspections, capture evidence, assign corrective actions, and identify risk trends across one or more mines.

The product will replace fragmented paper-based inspection and spreadsheet-driven compliance processes with a shared, auditable workflow. Its central value is not merely recording issues; it is creating accountability from requirement through inspection, finding, corrective action, verification, and closure. An AI-assisted risk engine will use operational data to produce mine-level risk scores and identify recurring violations, while a map-based view will help users understand where mines and issues are concentrated.

The first release should prioritize a reliable operational workflow over advanced artificial intelligence. The MVP must make it easy for an inspector to complete an inspection in the field, including photo, GPS, and timestamp capture, and must make it difficult for unresolved findings to disappear without ownership, evidence, and verification.

Product promise: Every compliance requirement, inspection observation, and corrective action should have a clear owner, deadline, status, evidence trail, and next action.

2. Problem Statement

Mining organizations often operate across multiple organizational levels and geographically distributed sites. Compliance information may be scattered across paper forms, spreadsheets, email, messaging applications, and isolated systems. This creates several

operational problems: managers lack a consolidated view of compliance health; inspectors spend time transcribing field notes; observations may not be linked to corrective actions; overdue actions are difficult to escalate; and recurring issues may remain invisible because historical records are not analyzed together.

The application addresses this problem by establishing a common data model and workflow for the organization, subsidiary, mine, and department hierarchy. It connects regulatory requirements to inspections, observations, corrective actions, evidence, verification, risk, maps, and alerts. The platform must support both field execution and management oversight without assuming that all users work at one mine.

The requirements in this document are derived from the user-provided concept brief ¹. Any regulatory rules, jurisdiction-specific forms, retention periods, or legal obligations must be confirmed with the customer before production deployment.

3. Product Vision and Goals

3.1 Vision

Create a trusted digital system of record for mining compliance and safety operations that helps organizations detect risk earlier, close issues faster, and demonstrate accountable compliance with an auditable evidence trail.

3.2 Product goals

Goal	Desired outcome	Primary measurement
Digitize field inspections	Inspectors can complete inspections without paper transcription	At least 90% of pilot inspections submitted digitally
Improve corrective-action accountability	Every material observation has an owner, deadline, and status	At least 95% of high-severity observations assigned within one business day
Increase compliance visibility	Managers can view compliance by organization, subsidiary, mine, and department	Dashboard loads with current status for all authorized mines
Reduce overdue work	Users receive timely reminders and escalations	Reduction in overdue corrective actions during pilot
Surface operational risk	Leaders can identify critical mines and recurring violation patterns	Risk score and recurring-pattern explanations available for eligible mines

Support defensible reporting	Evidence and change history are retained in a traceable record	100% of closed findings retain closure evidence and audit history
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3.3 Non-goals for MVP

The MVP will not attempt to replace a full enterprise resource planning system, perform autonomous regulatory interpretation, provide a complete geographic information system, or make final regulatory or safety decisions without human review. AI-generated risk indicators will be decision support, not a substitute for an inspector, mine manager, regulator, or safety professional.

4. Users, Roles, and Permissions

The product uses role-based access control combined with organizational scope. A user may be authorized for one mine, multiple mines, a subsidiary, an entire organization, or a regulatory portfolio. Permissions must be enforced in the mobile client and server-side APIs.

Role	Core responsibility	Typical permissions
Administrator	Configure and govern the platform	Manage users, roles, organizations, mines, departments, templates, settings, and audit access
Mine Manager	Manage operational compliance for an assigned mine	View mine dashboards, assign actions, review inspections, manage deadlines, and approve closure where authorized
Inspector	Conduct inspections and record evidence	View assigned mines, start inspections, create observations, capture media/GPS/time, submit reports, and update permitted records
Regulatory Authority	Monitor compliance across an authorized portfolio	View compliance, inspections, violations, corrective actions, evidence, risk, escalations, and reports; edit rights should be explicitly configurable

Corporate Management	Monitor multiple mines and subsidiaries	View cross-mine dashboards, compare trends, review risk, export reports, and receive escalations
Corrective-Action Owner	Resolve assigned findings	View assigned actions, submit progress, upload evidence, request verification, and respond to comments
Verifier or Approver	Confirm whether resolution is adequate	Review evidence, approve or reject resolution, add verification notes, and close or reopen the action

4.1 Access-control requirements

The system shall implement least-privilege access. A user must only see mines, departments, inspections, findings, evidence, and reports within their assigned scope unless an explicit cross-organization permission is granted. Sensitive evidence must inherit the permissions of its parent record. Every permission-sensitive action must be logged with the actor, timestamp, object, and result.

The system should support optional multi-factor authentication, configurable session expiry, password reset, account deactivation, and invitation-based onboarding. Authentication implementation may use an enterprise identity provider in later releases; MVP may begin with secure email-and-password authentication if approved by the customer.

5. Organization and Mine Management

5.1 Hierarchical data model

The product must represent a mining organization using the following hierarchy:

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Organization
├── Subsidiary
│   ├── Mine
│   │   └── Department

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A mine must not be modeled as the top-level entity. The design must support one organization with multiple subsidiaries, one subsidiary with multiple mines, and one mine with multiple departments. Users, compliance requirements, inspections, observations, actions, dashboards, and reports must be filterable at each relevant level.

5.2 Mine profile

Each mine profile should include name, code, organization, subsidiary, operating status, address, geographic coordinates, timezone, operational contacts, departments, risk summary, compliance summary, and audit metadata. The system should support a mine boundary or approximate operating area in a later release, but the MVP requires at least a map marker coordinate.

5.3 Department management

Authorized administrators and mine managers shall be able to create, edit, archive, and assign departments. Departments may include safety, electrical, operations, environment, equipment, human resources, or other customer-defined groups. Department ownership should be available for routing compliance requirements and corrective actions.

6. Compliance Management

A compliance requirement represents something a mine must do because of an internal policy, permit, regulation, standard, or other governing rule. The product shall maintain a structured compliance register rather than relying only on free-text notes.

6.1 Compliance requirement fields

Field	Description	Required for MVP
Requirement title	Human-readable obligation name	Yes
Requirement description	What must be done or demonstrated	Yes
Source or reference	Regulation, policy, permit, standard, or internal reference	Recommended
Applicable organization scope	Organization, subsidiary, mine, or department	Yes
Responsible role or user	Accountable person or role	Yes

Frequency	One-time, daily, weekly, monthly, quarterly, annual, or custom	Yes
Due date or schedule	Current and future due dates	Yes
Evidence type	Inspection, document, photo, measurement, approval, or custom	Recommended
Severity if missed	Business or safety impact of non-compliance	Recommended
Status	Pending, due soon, overdue, completed, or critical	Yes
Attachments and notes	Supporting information	Optional

6.2 Compliance statuses

The application shall support at least the following status categories: **Completed**, **Pending**, **Due Soon**, **Overdue**, and **Critical**. Status transitions must be calculated from due dates, completion state, severity, and escalation rules, while allowing authorized users to add a documented override reason.

The default configurable timing should classify an item as due soon when it falls within a configurable reminder window, such as 30 days before the due date. The customer must be able to configure this window by requirement type or organizational policy.

6.3 Compliance dashboard

The dashboard must show compliance performance for the user's permitted scope. It should provide overall completion percentage, counts by status, upcoming deadlines, overdue requirements, critical requirements, responsible owners, and trends over time. Users should be able to drill down from organization to subsidiary, mine, department, and individual requirement.

7. Inspection Management

Inspection management is the primary field workflow. An inspector should be able to move from assignment to submitted inspection with minimal typing and reliable evidence capture.

7.1 Inspection workflow

Plain Text

Select mine

- Select inspection template or type
- Start inspection
- Complete checklist and record observations
- Capture photo or other evidence
- Capture GPS and timestamp
- Review inspection
- Submit
- Generate findings and corrective actions where applicable

7.2 Inspection requirements

The mobile application shall allow an authorized inspector to:

1. View assigned mines and scheduled inspections.
2. Select a mine and inspection template.
3. Start, pause, resume, and submit an inspection.
4. Complete checklist items using configurable response types such as pass, fail, not applicable, numeric, text, or selection.
5. Create one or more observations from an inspection.
6. Record category, description, severity, location, and immediate action.
7. Capture photos from the device camera and attach existing images where permitted.
8. Capture device GPS coordinates and timestamp, subject to explicit user permission and local privacy requirements.
9. Continue in an offline mode when connectivity is unavailable and synchronize later.
10. Review a submission summary before final submission.
11. See synchronization state and resolve conflicts without losing local work.

7.3 Inspection templates

Administrators should be able to configure reusable inspection templates containing sections, checklist questions, response types, guidance text, required evidence rules, severity defaults, and applicable departments. A submitted inspection must retain the template version used at the time of completion so that later template changes do not alter historical records.

7.4 Inspection evidence

Each inspection and observation may contain photographs, comments, documents, measurements, and location metadata. Evidence must retain upload time, uploader, file type, parent record, and integrity metadata. The application should compress images for upload while retaining an original or high-quality version when required by policy.

8. Violation and Observation Management

An observation is a finding recorded during an inspection or created through an authorized follow-up workflow. Observations must be searchable, traceable to their source inspection, and linked to one or more corrective actions when resolution is required.

8.1 Observation fields

Field	Description
Observation identifier	System-generated unique number
Issue description	Clear statement of the observed condition
Mine and department	Operational location
Category	Safety, electrical, equipment, environment, process, or customer-defined category
Severity	Configurable scale, at minimum low, medium, high, and critical
Status	Open, assigned, in progress, resolved, verified, or closed
Inspector and source inspection	Originating user and inspection reference
Location	GPS coordinates and optional descriptive location
Date and time	Observation timestamp
Evidence	Photos, documents, comments, and measurements
Immediate containment	Action taken to reduce immediate harm, if applicable
Corrective action link	Linked action record and owner

8.2 Severity and criticality

Severity must be configurable by the customer. Critical observations should trigger immediate escalation according to policy and may require an immediate containment record before the inspection can be submitted. Severity changes must be auditable and may require an authorized approver.

8.3 Search, filtering, and history

Users should be able to search and filter observations by mine, department, category, severity, status, date range, inspector, owner, recurring pattern, and overdue state. The detail view must show the complete history of changes, comments, evidence, assignments, deadlines, and verification decisions.

9. Corrective-Action Management

Corrective actions convert findings into accountable work. A corrective action must include an owner, due date, expected resolution, and status progression.

9.1 Corrective-action lifecycle

Plain Text

OPEN

- ASSIGNED
- IN PROGRESS
- RESOLVED
- VERIFIED
- CLOSED

An authorized user may return an action to an earlier state when evidence is insufficient or the issue reopens. Reopening must require a reason. The system must not allow a corrective action to be marked closed without an authorized verification step unless an administrator-approved exception is recorded.

9.2 Corrective-action fields

Field	Description	Required
Action identifier	Unique system-generated number	Yes
Linked observation	Source finding	Yes

Action description	Specific remediation required	Yes
Owner	Person accountable for completion	Yes
Reviewer or verifier	Person accountable for verification	Recommended
Priority	Derived from observation severity or manually configured	Yes
Due date	Deadline for resolution	Yes
Status	Lifecycle state	Yes
Progress notes	Ongoing updates	Optional
Resolution evidence	Photos, documents, or measurements	Required before resolution
Verification decision	Accept, reject, or request changes	Required before closure
Escalation state	Normal, warned, escalated, or critical	Recommended

9.3 Verification workflow

When an owner submits an action as resolved, the designated verifier receives a notification. The verifier can approve and close the action, reject the evidence with comments, or request additional evidence. The owner must be able to see the reason for rejection and submit a new resolution package.

10. AI Risk Engine

The AI risk engine will provide an explainable risk score for a mine based on operational signals. The MVP should begin with a transparent weighted scoring model that can later be enhanced with machine-learning methods after sufficient historical data is available.

10.1 Initial risk inputs

The initial model should consider violation frequency, violation severity, overdue corrective actions, historical incidents, inspection performance, compliance completion, and recurrence of the same or related issue categories. Weights must be configurable by an administrator or product configuration, with defaults documented and versioned.

10.2 Risk output

The engine should produce:

Output	Requirement
Numeric risk score	A normalized score, for example 0–100
Risk level	Low, moderate, high, or critical
Contributing factors	Ranked explanation of the main drivers
Comparison period	Date range used for the calculation
Confidence or data sufficiency	Indicates whether enough data exists for a meaningful score
Recommended attention	Human-readable next-step prompt
Model version	Version of scoring logic used

Example output: “Mine A has a risk score of 84, classified as Critical. The principal contributors are 8 overdue actions, 15 violations, 3 incidents, and an inspection performance score of 62%.” The application must present this as an indicator requiring human review rather than as an unquestionable conclusion.

10.3 Risk governance

Risk scores must be reproducible from stored inputs, explainable to authorized users, and recalculated when material source data changes or on a scheduled basis. The system must preserve the score history and model version. Users must be able to identify missing data and challenge or annotate an assessment.

11. Recurring Violation Detection

The application should identify repeated violations that may indicate a systemic problem. The MVP can use deterministic rules based on category, tags, mine, department, and time window before introducing advanced semantic models.

For example, if PPE-related observations occur in January, March, May, and July, the system should generate an insight such as: “PPE violations have occurred four times in the last seven months.” The insight should include the records used, the time window, the matching category or tags, and a link to the relevant observations.

Recurring-pattern requirements include configurable thresholds, suppression of duplicate alerts, user acknowledgment, severity-aware prioritization, and the ability to mark a pattern as investigated. AI-generated descriptions must not alter source records automatically.

12. GIS and Map Experience

The map view will provide a lightweight geographic overview rather than a full enterprise GIS. The MVP must show mine markers, risk indicators, and optionally issue markers when location data is available.

Selecting a mine marker should open a summary containing compliance percentage, risk score and level, open violations, overdue corrective actions, and a link to the mine detail page. Marker color should communicate risk category, with a clear legend and accessible non-color labels.

The map must support permission-based filtering, search by mine, clustering when many markers overlap, and a list alternative for users who cannot or do not wish to use a map. GPS coordinates captured during inspections should be displayed only to users with appropriate access.

13. Alerts, Reminders, and Escalation

The system shall notify relevant users when compliance or corrective-action deadlines approach or are missed. Notification channels should include in-app notifications and push notifications in the mobile app. Email and SMS may be added based on customer requirements and service availability.

13.1 Default escalation schedule

Trigger	Default notification	Recipients
30 days before due date	Reminder	Responsible owner and mine manager
7 days before due date	Warning	Responsible owner and mine manager
Due date	Due today	Responsible owner, mine manager, and configured escalation role

Deadline passed	Overdue escalation	Owner, mine manager, and configured escalation role
Critical observation	Immediate escalation	Mine manager, safety leadership, and other configured recipients

All thresholds, recipients, quiet hours, timezones, and escalation chains must be configurable. Notifications must be idempotent so a synchronization retry cannot create duplicates. Each notification must link directly to the relevant compliance requirement, observation, or corrective action.

14. Mobile Application Experience

The mobile app should be designed for field conditions: intermittent connectivity, bright outdoor environments, gloves or limited dexterity, time pressure, and the need to capture evidence quickly. The primary navigation should expose Home, Inspections, Observations, Actions, Map, Notifications, and Profile or Settings.

14.1 Essential screens

Screen	Purpose
Sign in and onboarding	Authenticate users and establish authorized scope
Home dashboard	Show assigned work, overdue items, critical alerts, and quick actions
Mine selector	Select an authorized mine and view its summary
Inspection list	Show scheduled, draft, submitted, and synchronized inspections
Inspection execution	Complete checklist items and capture findings
Observation form	Record issue, severity, category, evidence, and location
Observation detail	Review history, evidence, and linked actions
Corrective-action detail	Update progress, upload evidence, and request verification

Verification view	Approve, reject, or request changes to resolution evidence
Risk dashboard	Show score, level, drivers, trend, and recurring patterns
Map view	Explore mines and issue indicators geographically
Notifications	Review reminders, warnings, escalations, and acknowledgments
Profile and settings	Manage account, permissions visibility, notification preferences, and offline sync

14.2 Offline mode

The app shall support creation and editing of inspections, observations, and evidence while offline. Data must be encrypted locally and synchronized when connectivity returns. The app must clearly show whether records are local, queued, synchronized, or failed. Conflicts must be surfaced to the user with a safe resolution path; silent overwriting is not acceptable.

15. Reporting and Auditability

The system should provide dashboards and exports for operational, management, and regulatory audiences. MVP reports should include compliance status, inspection activity, open observations, overdue corrective actions, closure performance, risk by mine, recurring violations, and escalation history.

Reports must respect authorization scope and show the reporting period, generated time, filters, and data freshness. Exports should be available in CSV and PDF only if required by the customer; the default MVP can prioritize CSV and an in-app report view.

An immutable or append-only audit trail must record significant actions, including sign-in events where required, record creation, updates, status changes, assignment changes, evidence uploads and deletions, verification decisions, risk-score generation, permission changes, and exports. Audit records should include actor, timestamp, action, object, previous value where appropriate, new value, and source device or session metadata where legally permissible.

16. Non-Functional Requirements

Area	Requirement
Availability	The production service should target a documented availability objective appropriate to the customer' s operating model.
Performance	Common dashboard and detail views should load within an agreed target under normal network conditions; the target should be validated during discovery.
Security	Data must be encrypted in transit and at rest. Access must be server-authorized, not trusted solely because a mobile client hides a screen.
Privacy	The product must collect only required location, identity, and evidence data and must provide customer-configurable retention and access controls.
Reliability	Offline records must survive app restarts and synchronization failures without data loss.
Scalability	The data model must support multiple organizations, subsidiaries, mines, departments, users, and inspection records.
Accessibility	Text, controls, status labels, and map alternatives should support accessible use and must not communicate meaning by color alone.
Localization	Dates, times, number formats, timezones, and future language support should be configurable.
Observability	The service should provide error logging, synchronization metrics, notification delivery status, and audit monitoring.
Maintainability	Compliance templates, severity scales, escalation thresholds, and risk weights should be configurable without a mobile release where practical.

Final security, privacy, retention, availability, and regulatory requirements must be confirmed during customer discovery because they may vary by jurisdiction and

deployment model.

17. Recommended Free Technology Stack

The MVP should use a React-based stack and the customer’ s existing **Supabase** project. This minimizes infrastructure cost and development time while providing authentication, relational data, file storage, real-time updates, and server-side functions in one platform. The free tier is suitable for prototyping and a controlled pilot; production deployment must be checked against current Supabase, app-store, map-tile, and notification limits before launch.

17.1 Stack selection

Layer	Recommended technology	Responsibility
Mobile application	React Native with Expo and TypeScript	Cross-platform Android and iOS field application
Optional management web app	React with Vite and TypeScript	Corporate, regulatory, and administrator dashboards
UI and styling	NativeWind or React Native StyleSheet; Tailwind CSS for web	Consistent responsive interface
Client state	TanStack Query plus Zustand	Server cache, synchronization state, and local UI state
Forms and validation	React Hook Form with Zod	Inspection forms, observations, and corrective-action validation
Authentication	Supabase Auth	Sign-in, sessions, password reset, invitations, and future SSO integration
Database	Supabase PostgreSQL	Multi-tenant organizational, compliance, inspection, action, and audit data
Authorization	PostgreSQL Row Level Security policies	Server-enforced organization, subsidiary, mine, and role access
File storage	Supabase Storage	Photos, documents, inspection evidence, and resolution evidence

Backend logic	Supabase Edge Functions	Notifications, scheduled calculations, privileged operations, and integrations
Offline data	expo-sqlite or an encrypted local store	Draft inspections, queued evidence, and synchronization state
Maps	MapLibre React Native or Leaflet for web	Mine markers, issue markers, and risk visualization
Map data	OpenStreetMap-compatible provider	Basemap tiles; provider terms and usage limits must be reviewed
Push notifications	Expo Notifications for mobile; email through an approved provider if required	Reminders, warnings, due-date alerts, and escalations
Hosting	Expo development/build services for the mobile app; Vercel or Netlify for the optional web app	Preview and production delivery
Testing	Jest, React Native Testing Library, and Playwright for web	Unit, workflow, accessibility, and regression testing
Source control	GitHub	Code review, issue tracking, and release history

17.2 Supabase data architecture

The Supabase database should use a shared multi-tenant schema with explicit organization scope on all business records. Core tables should include `organizations` , `subsidiaries` , `mines` , `departments` , `profiles` , `roles` , `user_scopes` , `compliance_requirements` , `inspection_templates` , `inspection_template_versions` , `inspections` , `inspection_responses` , `observations` , `corrective_actions` , `action_status_history` , `evidence_items` , `risk_assessments` , `recurring_patterns` , `notifications` , and `audit_events` .

Every table containing customer data must include the appropriate organization or mine relationship. Row Level Security policies must be enabled and tested for every table, view, storage bucket, and database function. The application must never rely only on React route guards for authorization because a user could bypass client-side controls by calling the API directly.

Supabase Storage should use private buckets for inspection and corrective-action evidence. Files should be accessed through short-lived signed URLs after server-side authorization. Public buckets should not be used for safety, employee, incident, or regulatory evidence.

17.3 React mobile architecture

The mobile application should provide role-aware navigation, an offline inspection workflow, a local synchronization queue, camera and GPS permissions, image compression, retryable uploads, and clear synchronization states. A recommended flow is:

Plain Text

React Native + Expo Mobile App

- |— Supabase Auth session
- |— TanStack Query data cache
- |— expo-sqlite offline store
- |— Sync queue and conflict handling
- |— Inspection, observation, and action screens
- |— Camera, GPS, and timestamp capture
- |— Map and notification features



Supabase

- |— PostgreSQL database with Row Level Security
- |— Supabase Auth
- |— Private Storage buckets
- |— Realtime subscriptions where useful
- |— Edge Functions for privileged or scheduled work

17.4 Risk and AI implementation

The first version of the risk engine should be implemented as a transparent weighted calculation in PostgreSQL or a Supabase Edge Function. It should calculate the score from violation frequency, severity, overdue actions, incidents, inspection performance, compliance status, and recurring patterns. The calculation must store its inputs, weights, result, explanation, calculation period, and model version.

Advanced AI should be optional and introduced later through a Supabase Edge Function. No secret AI key may be placed in the React or React Native client. AI output must remain advisory, explainable, versioned, and subject to human review. If no paid AI service is available, the MVP can provide recurring-violation detection using SQL rules, categories, tags, and configurable time windows.

17.5 Free-stack constraints

This stack is appropriate for an MVP and pilot, but “free” does not mean unlimited or automatically production-ready. The project must monitor database size, storage usage, bandwidth, function invocations, authentication email delivery, map-tile usage, notification limits, build limits, and app-store distribution costs. The team should also maintain a backup and export procedure because a free hosted database must not be treated as the only copy of critical compliance evidence.

Before production, the team must confirm data residency, retention, backups, disaster recovery, privacy obligations, map-provider terms, and whether the expected inspection volume fits the selected free service tiers.

18. Core Data Entities

Entity	Key relationships
User	Belongs to organization; has roles and scope assignments
Organization	Contains subsidiaries
Subsidiary	Belongs to organization; contains mines
Mine	Belongs to subsidiary; contains departments, inspections, findings, and actions
Department	Belongs to mine; may own requirements and actions
Compliance requirement	Applies to organizational scope; has owner, schedule, evidence, and status
Inspection template	Contains versioned sections and checklist items
Inspection	Uses a template; belongs to mine; created by inspector; contains responses and observations
Observation	Belongs to inspection and mine; may create corrective actions
Corrective action	Belongs to observation; has owner, verifier, deadline, evidence, and status history
Evidence item	Belongs to inspection, observation, or action; stored in object storage

Risk assessment	Belongs to mine and period; stores score, level, drivers, and model version
Recurring pattern	Links related observations and stores detection rule, period, and status
Notification	Links recipient, trigger, target record, delivery state, and acknowledgment
Audit event	Records actor, action, object, time, and change details

19. MVP Scope and Release Plan

19.1 MVP release

The MVP should include secure authentication, role-aware access, organization and mine hierarchy, mine profiles, compliance requirements, inspection templates, offline-capable inspection execution, photo/GPS/timestamp evidence capture, observations, corrective actions, verification, basic dashboards, configurable alerts, a basic mine map, audit history, and an explainable weighted risk score.

19.2 Release 2

The second release may add advanced recurring-violation detection, richer reporting and exports, email or SMS escalation, more robust template administration, bulk import, stronger regulatory portfolio views, map overlays, advanced analytics, and enterprise identity-provider integration.

19.3 Future capabilities

Future releases may introduce semantic similarity for related findings, predictive risk modeling, automated document extraction, voice-assisted inspection entry, integration with incident or asset systems, advanced geospatial layers, and configurable regulatory packs. Such capabilities should be introduced only after data quality, consent, governance, and human-review controls are established.

20. Acceptance Criteria for MVP

The MVP will be considered functionally complete when an authorized administrator can create an organization, subsidiary, mine, department, users, roles, and inspection

template; an inspector can select an authorized mine, complete an inspection offline, create an observation, capture photo/GPS/timestamp evidence, and synchronize the submission; and the system can create a corrective action with an owner and deadline.

It must also be possible for the owner to update the action, upload resolution evidence, request verification, and for an authorized verifier to approve or reject the resolution. The system must calculate compliance and risk summaries, display mines on a map, detect and notify overdue work, and preserve an audit trail of the material workflow. A user must never be able to access records outside the user's permitted scope through a direct API request.

21. Analytics and Success Metrics

Metric	Definition	Initial target or use
Digital inspection completion	Submitted inspections completed in the app divided by total pilot inspections	Track adoption; target to be set with pilot customer
Inspection submission time	Median time from inspection start to submission	Establish baseline and reduce transcription burden
Observation assignment time	Time from observation creation to owner assignment	Monitor accountability; target under one business day for high severity
Corrective-action closure time	Time from action creation to verified closure	Track operational improvement by mine and severity
Overdue action rate	Overdue open actions divided by all open actions	Reduce over time through reminders and escalation
Verification rejection rate	Resolutions rejected by verifier divided by submitted resolutions	Identify quality or training gaps
Evidence completeness	Findings with required evidence divided by findings requiring evidence	Improve audit readiness
Recurring-violation acknowledgment	Detected patterns reviewed or acknowledged by authorized users	Measure usefulness of AI insights
Risk review engagement	Mine risk assessments viewed or acted upon	Measure management adoption

Synchronization reliability	Successful sync operations divided by attempted operations	Ensure field usability
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Product analytics must avoid collecting unnecessary personal information. Customer-facing dashboards should distinguish between system activity metrics and safety or compliance outcomes.

22. Risks and Mitigations

Risk	Impact	Mitigation
Poor field connectivity	Inspections cannot be completed reliably	Offline-first workflows, durable local queue, visible sync state, and conflict handling
Inconsistent data entry	Risk and recurring-pattern results become unreliable	Templates, required fields, validation, controlled categories, and training
AI over-trust	Users treat a score as a final safety decision	Explainable factors, confidence/data sufficiency, human review, and clear disclaimers
Excessive notifications	Users ignore important escalations	Configurable rules, deduplication, priority channels, and user preferences
Unauthorized access	Sensitive operational and safety data is exposed	Server-side authorization, least privilege, audit logs, and security testing
Closure without real resolution	Compliance records become misleading	Mandatory evidence and independent verification for applicable severities
Scope expansion	Delivery is delayed by trying to build a complete GIS or regulatory suite	Protect MVP boundaries and use staged releases
Jurisdictional differences	Workflows may not fit local requirements	Configuration, discovery with domain experts, and legal/compliance review

23. Open Questions for Discovery

The following questions must be answered before final design and estimation:

1. Which countries, jurisdictions, mining types, and regulatory frameworks are in scope for the pilot?
 2. Is the product intended for internal corporate use, regulator use, or a shared multi-tenant ecosystem?
 3. What identity provider, device-management policy, and data-residency requirements apply?
 4. Which users may create, assign, verify, reopen, or close corrective actions?
 5. What severity taxonomy and escalation rules are already used by the customer?
 6. Which evidence types are legally or operationally required for each compliance category?
 7. How long must inspections, images, audit events, and closed actions be retained?
 8. Which map provider and coordinate accuracy are acceptable for the pilot?
 9. What happens when an inspector captures evidence without location permission or with inaccurate GPS?
 10. Are there existing systems for incidents, assets, HR, permits, or regulatory reporting that must be integrated?
 11. What are the expected user counts, mine counts, inspection volume, and peak synchronization patterns?
 12. Which languages, accessibility needs, and device classes must be supported?
 13. What is the approved definition of an “incident” for the risk calculation?
 14. Which decisions must remain exclusively human-controlled?
 15. What operational baseline will be used to measure the product’s impact?
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24. Recommended Delivery Phases

Phase	Focus	Exit criteria
Discovery	Validate users, workflows, regulatory context, data, and technical constraints	Approved scope, personas, process maps, and prioritized backlog

Foundation	Authentication, organization hierarchy, roles, mine profiles, data model, and audit foundation	Users can securely access authorized organizational scope
Field workflow	Inspection templates, offline execution, observations, evidence capture, and synchronization	Pilot inspector can complete and submit a field inspection
Accountability	Corrective actions, notifications, verification, dashboards, and reporting	Finding-to-closure workflow works end to end
Intelligence	Risk score, recurring detection, map indicators, and insight explanations	Managers can review risk drivers and recurring issues
Pilot hardening	Security, performance, device testing, training, analytics, and operational support	Pilot readiness sign-off and measured rollout plan

25. Final Product Principle

The product should always optimize for trustworthy action rather than attractive data display. A risk score without explanation, an observation without an owner, a corrective action without verification, or a dashboard without current data is not a complete compliance solution. Every major feature should therefore answer four questions: **What happened? Who is responsible? What evidence exists? What must happen next?**

References

[1] User-provided concept brief: AI-based mining compliance, inspection, risk, GIS, and alerts modules

This PRD is a product-planning document, not legal or regulatory advice. Jurisdiction-specific requirements should be validated with qualified customer representatives and relevant compliance professionals before implementation.